PyPSA-NL Grid Flexibility Report

Human-readable executive report with charts, tables and embedded full data bundle.

Generated on 2026-05-03 06:01:36 | The attached ZIP bundle contains the complete machine-readable inputs, outputs, figures and reports.

Latest key findings

Top-ranked scenario: High Wind Offshore Growth with grid-value score 60.8.

It adds 201,561 MWh of renewable dispatch and reduces backup generation by 201,561 MWh versus base.

Congestion-cost proxy is reduced by EUR 7,984,655 versus base.

Absolute curtailment is higher than base by 170,594 MWh, so the ranking should be read as a multi-KPI trade-off rather than a curtailment-only result.

Base case curtailment is 484,638 MWh at 24.4%.

Best BESS sweep option: Groningen 400 MW / 4 h, with score 8.9.

Validation checks passed: 8/8.

Scenario ranking snapshot

Scenario	Score	Renewable gain [MWh]	Backup reduction [MWh]	Curtailment [%]	Congestion proxy [EUR]	Rank
high_wind_offshore_growth	60.85	201,560.84	201,560.84	27.82	104,219,193.54	1.00
combined_bess_and_flex	30.68	186,689.75	146,072.94	35.25	127,412,360.84	2.00
targeted_grid_reinforcement	26.14	110,563.36	110,563.36	31.61	118,432,707.50	3.00
bess_noord_holland	23.13	114,786.74	114,093.29	31.43	118,572,844.10	4.00
bess_noord_brabant	22.41	114,786.74	114,093.29	31.43	118,692,844.10	5.00
bess_flevoland	19.89	114,852.77	114,106.62	31.43	119,107,285.68	6.00
base_2026_constrained_grid	0.00	0.00	0.00	24.44	112,203,848.15	7.00
flexible_connection_contract	-4.07	118,176.74	78,426.74	34.95	132,841,683.16	8.00
high_solar_growth	-19.54	30,057.32	30,057.32	37.59	139,196,464.78	9.00

Base-case reference metrics

Base curtailment [MWh]	Base curtailment rate [%]	Base backup [MWh]	Base congestion cost proxy [EUR]	Base renewable share [%]
484,638.39	24.44	727,130.03	112,203,848.15	67.32

Validation summary

Check	Pass	Value	Tolerance	Details
required_columns_present	1.00	none	none missing	Required KPI columns are available for validation.
curtailment_balance	1.00	4.656612873077393e-10	0.001	renewable_available_mwh - renewable_dispatch_mwh equals renewable_curtailment_mwh.
renewable_share_not_above_100_pct	1.00	76.38126863589343	<=100%	Renewable dispatch share of demand remains physically bounded.
curtailment_rate_between_0_and_100_pct	1.00	24.443 to 37.5871	0% to 100%	Curtailment rates are within expected percentage bounds.
bess_cycles_finite_non_negative	1.00	0	0	BESS equivalent cycles are finite and non-negative.
objective_cost_finite_non_negative	1.00	0	0	Objective costs are finite and non-negative.
all_scenarios_solved_optimal	1.00	all optimal	all optimal	Every scenario reports PyPSA status ok and optimal termination.
recommendation_ranks_complete	1.00	9	9	Recommendation ranks form a complete 1..N sequence.

Top BESS siting and sizing options

Region	Power [MW]	Duration [h]	Energy [MWh]	Score	Renewable gain [MWh]	Backup reduction [MWh]	Congestion relief [EUR]
Groningen	400.00	4.00	1,600.00	8.85	12,105.10	10,617.27	1,900,471.90
Zuid-Holland	400.00	4.00	1,600.00	7.93	12,110.92	10,605.99	1,809,778.48
Zeeland	400.00	4.00	1,600.00	6.63	11,936.37	10,546.98	1,672,827.05
Noord-Brabant	400.00	4.00	1,600.00	6.31	11,914.89	10,528.00	1,639,628.09
Flevoland	400.00	4.00	1,600.00	4.46	12,151.86	10,704.77	1,476,824.11
Noord-Brabant	200.00	4.00	800.00	4.30	5,957.45	5,264.00	932,314.04
Noord-Holland	400.00	4.00	1,600.00	4.29	12,137.42	10,660.19	1,456,650.35
Zuid-Holland	200.00	4.00	800.00	3.43	6,055.46	5,302.99	852,389.24
Zuid-Holland	400.00	2.00	800.00	3.31	6,375.42	5,530.21	864,772.55
Noord-Brabant	400.00	2.00	800.00	2.62	5,957.45	5,264.00	767,314.04
Zeeland	400.00	2.00	800.00	2.56	6,343.72	5,605.31	794,846.45
Groningen	400.00	2.00	800.00	2.35	6,308.21	5,473.78	765,042.23
Noord-Brabant	200.00	2.00	400.00	1.92	2,978.72	2,632.00	443,657.02
Zuid-Holland	400.00	1.00	400.00	1.82	3,320.50	2,874.71	458,930.10
Noord-Brabant	100.00	4.00	400.00	1.61	2,978.72	2,632.00	413,657.02

Bottleneck diagnostics snapshot

Scenario	Line	Max util. [%]	Hours > threshold	Severity [%h]	Score
bess_flevoland	Flevoland__Overijssel	100.00	122.00	1,176.57	1,233.73
flexible_connection_contract	Flevoland__Overijssel	100.00	112.00	1,067.96	1,120.39
high_wind_offshore_growth	Noord-Brabant__Zeeland	100.00	110.00	1,051.90	1,114.35
combined_bess_and_flex	Flevoland__Overijssel	100.00	102.00	994.11	1,041.95

Scenario	Line	Max util. [%]	Hours > threshold	Severity [%h]	Score
bess_noord_brabant	Flevoland__Overijssel	100.00	104.00	988.75	1,037.41
high_wind_offshore_growth	Flevoland__Overijssel	100.00	100.00	987.48	1,034.43
targeted_grid_reinforcement	Noord-Brabant__Zeeland	100.00	102.00	975.31	1,033.22
bess_noord_holland	Flevoland__Overijssel	100.00	102.00	970.13	1,017.86
high_wind_offshore_growth	Gelderland__Overijssel	100.00	98.00	927.63	991.02
high_solar_growth	Gelderland__Overijssel	100.00	97.00	923.66	986.43
bess_flevoland	Noord-Brabant__Zeeland	100.00	97.00	921.44	976.48
flexible_connection_contract	Friesland__Groningen	100.00	92.00	919.34	969.94

N-1 screening proxy snapshot

Scenario	Outaged line	Max util. [%]	Hours > threshold	Score	Interpretation
high_wind_offshore_growth	Gelderland__Overijssel	100.00	98.00	121,871.55	High-priority outage screening corridor
high_solar_growth	Gelderland__Overijssel	100.00	97.00	120,693.92	High-priority outage screening corridor
high_wind_offshore_growth	Noord-Brabant__Zeeland	100.00	110.00	119,399.73	High-priority outage screening corridor
combined_bess_and_flex	Gelderland__Overijssel	100.00	92.00	114,649.41	High-priority outage screening corridor
combined_bess_and_flex	Noord-Brabant__Limburg	100.00	86.00	110,737.43	High-priority outage screening corridor
targeted_grid_reinforcement	Noord-Brabant__Zeeland	100.00	102.00	110,715.24	High-priority outage screening corridor
bess_noord_holland	Noord-Brabant__Limburg	100.00	86.00	110,074.39	High-priority outage screening corridor
bess_flevoland	Flevoland__Overijssel	100.00	122.00	108,215.17	High-priority outage screening corridor
flexible_connection_contract	Noord-Brabant__Limburg	100.00	84.00	107,927.66	High-priority outage screening corridor
high_solar_growth	Noord-Brabant__Limburg	100.00	84.00	107,585.00	High-priority outage screening corridor
bess_flevoland	Gelderland__Overijssel	100.00	87.00	107,545.98	High-priority outage screening corridor
bess_noord_brabant	Gelderland__Overijssel	100.00	86.00	107,230.31	High-priority outage screening corridor

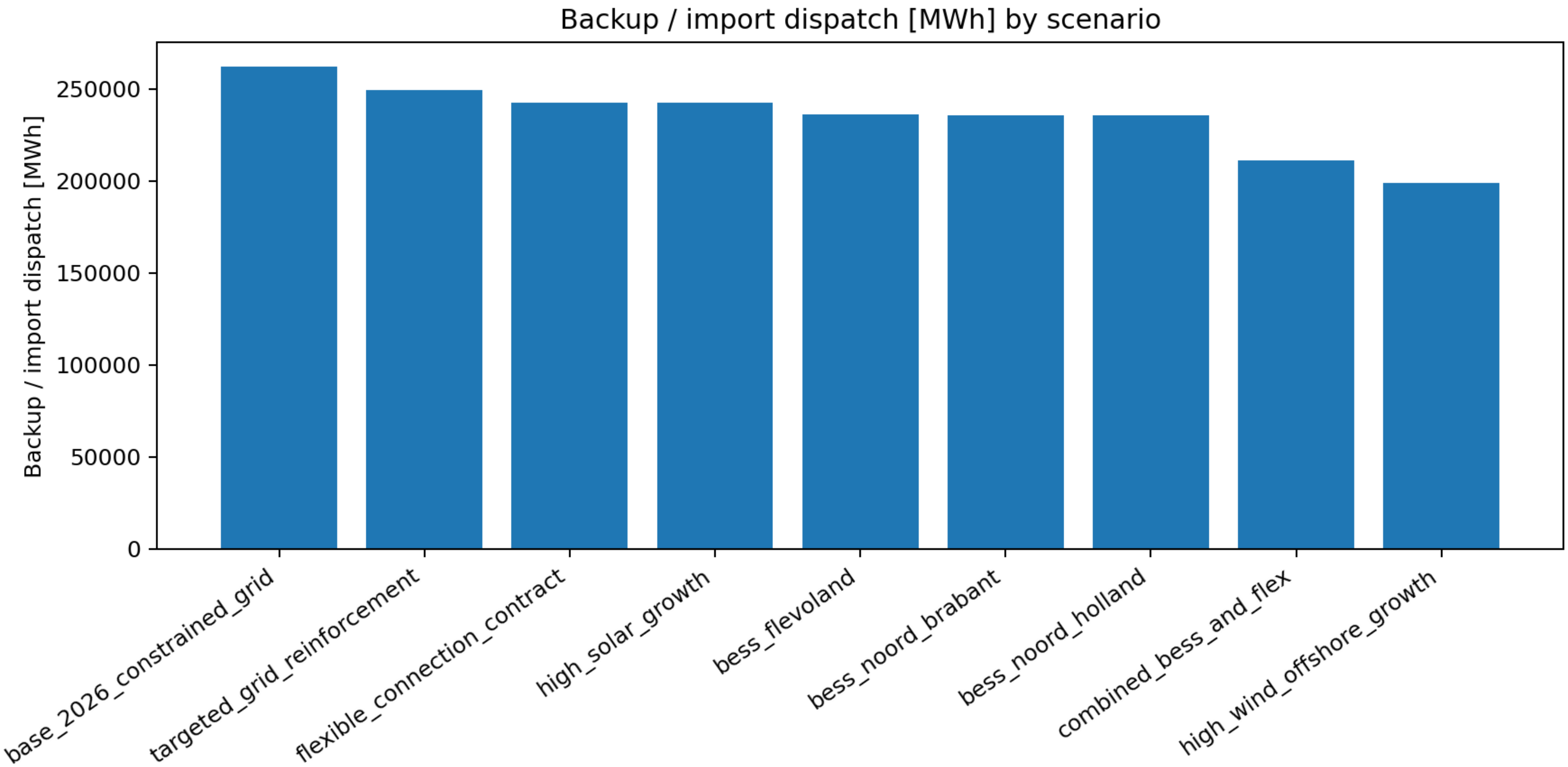
BESS business-case proxy snapshot

Region	Power [MW]	Duration [h]	CAPEX [EUR]	Annual cost [EUR/yr]	Annual value [EUR/yr]	Net value [EUR/yr]	BCR	Payback [yr]
Groningen	400.00	4.00	398,000,000.00	53,648,260.63	99,096,034.71	45,447,774.08	1.85	4.02
Zuid-Holland	400.00	4.00	398,000,000.00	53,648,260.63	94,367,020.53	40,718,759.90	1.76	4.22
Zeeland	400.00	4.00	398,000,000.00	53,648,260.63	87,225,981.73	33,577,721.10	1.63	4.56
Noord-Brabant	400.00	4.00	398,000,000.00	53,648,260.63	85,494,893.01	31,846,632.38	1.59	4.66
Flevoland	400.00	4.00	398,000,000.00	53,648,260.63	77,005,828.64	23,357,568.01	1.44	5.17
Noord-Holland	400.00	4.00	398,000,000.00	53,648,260.63	75,953,911.13	22,305,650.50	1.42	5.24
Noord-Brabant	200.00	4.00	199,000,000.00	26,824,130.32	48,613,517.93	21,789,387.62	1.81	4.09
Zuid-Holland	200.00	4.00	199,000,000.00	26,824,130.32	44,446,010.27	17,621,879.95	1.66	4.48

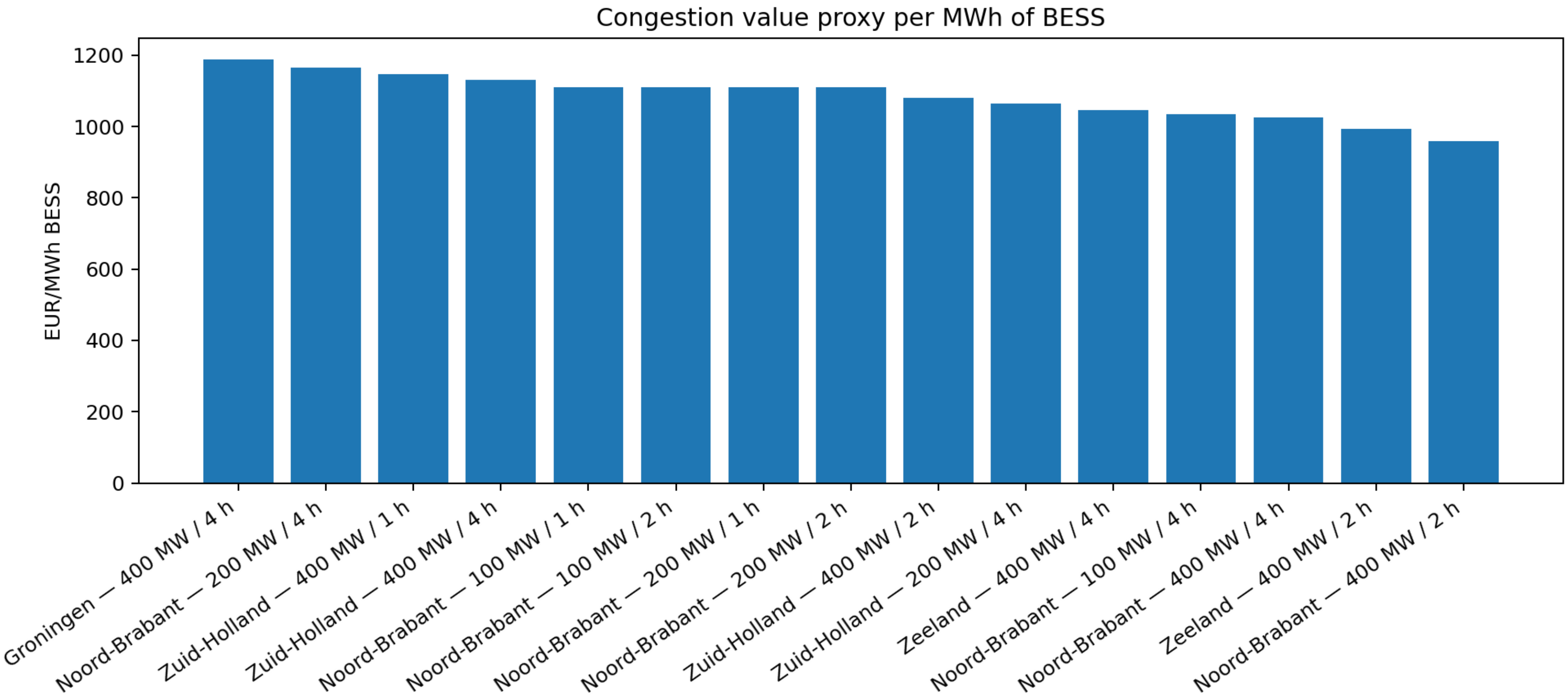
Region	Power [MW]	Duration [h]	CAPEX [EUR]	Annual cost [EUR/yr]	Annual value [EUR/yr]	Net value [EUR/yr]	BCR	Payback [yr]
Zuid-Holland	400.00	2.00	230,000,000.00	31,002,763.68	45,091,711.28	14,088,947.60	1.45	5.10
Zeeland	400.00	2.00	230,000,000.00	31,002,763.68	41,445,564.94	10,442,801.26	1.34	5.55
Noord-Brabant	400.00	2.00	230,000,000.00	31,002,763.68	40,009,946.50	9,007,182.82	1.29	5.75
Groningen	400.00	2.00	230,000,000.00	31,002,763.68	39,891,487.82	8,888,724.14	1.29	5.77

Figures

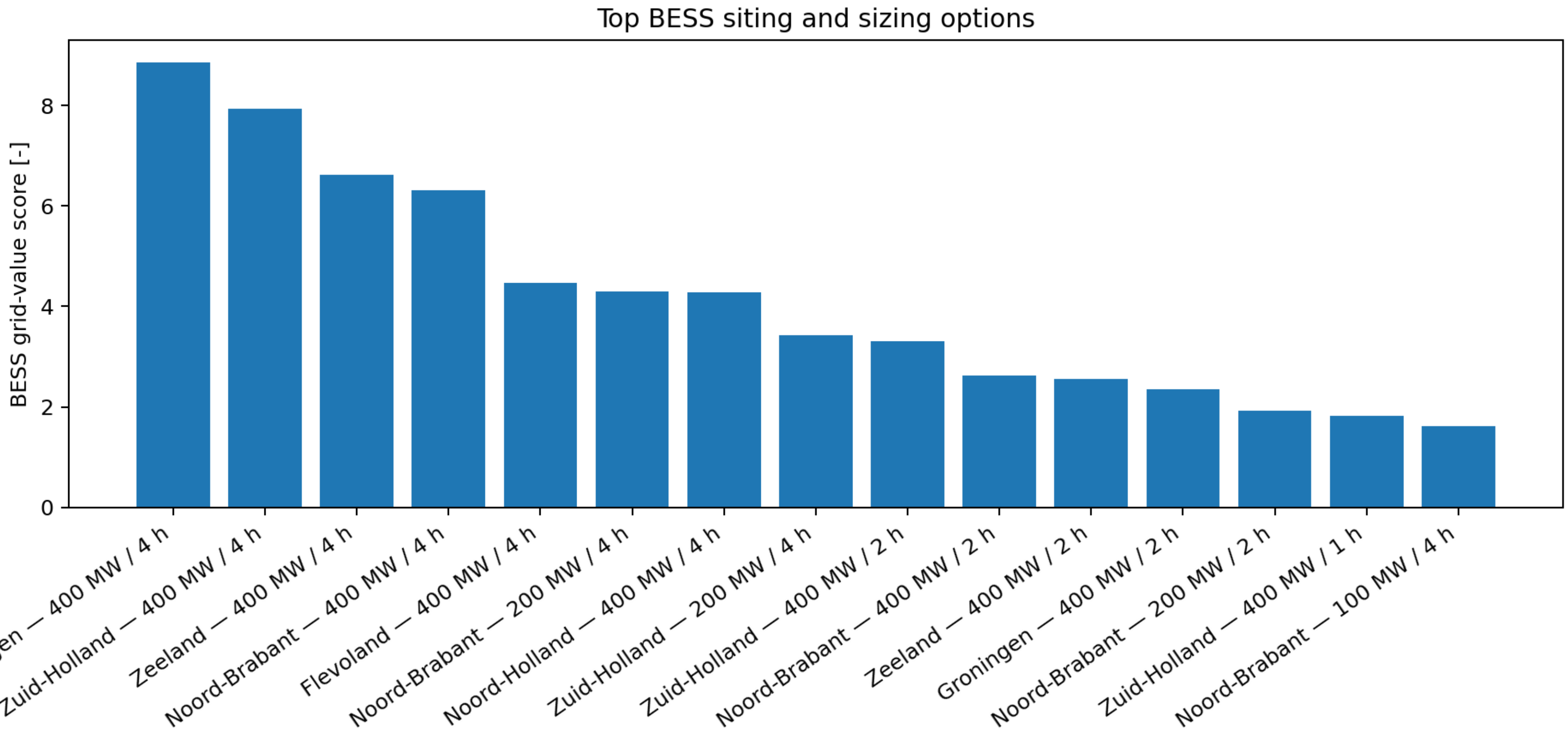
Backup Dispatch By Scenario



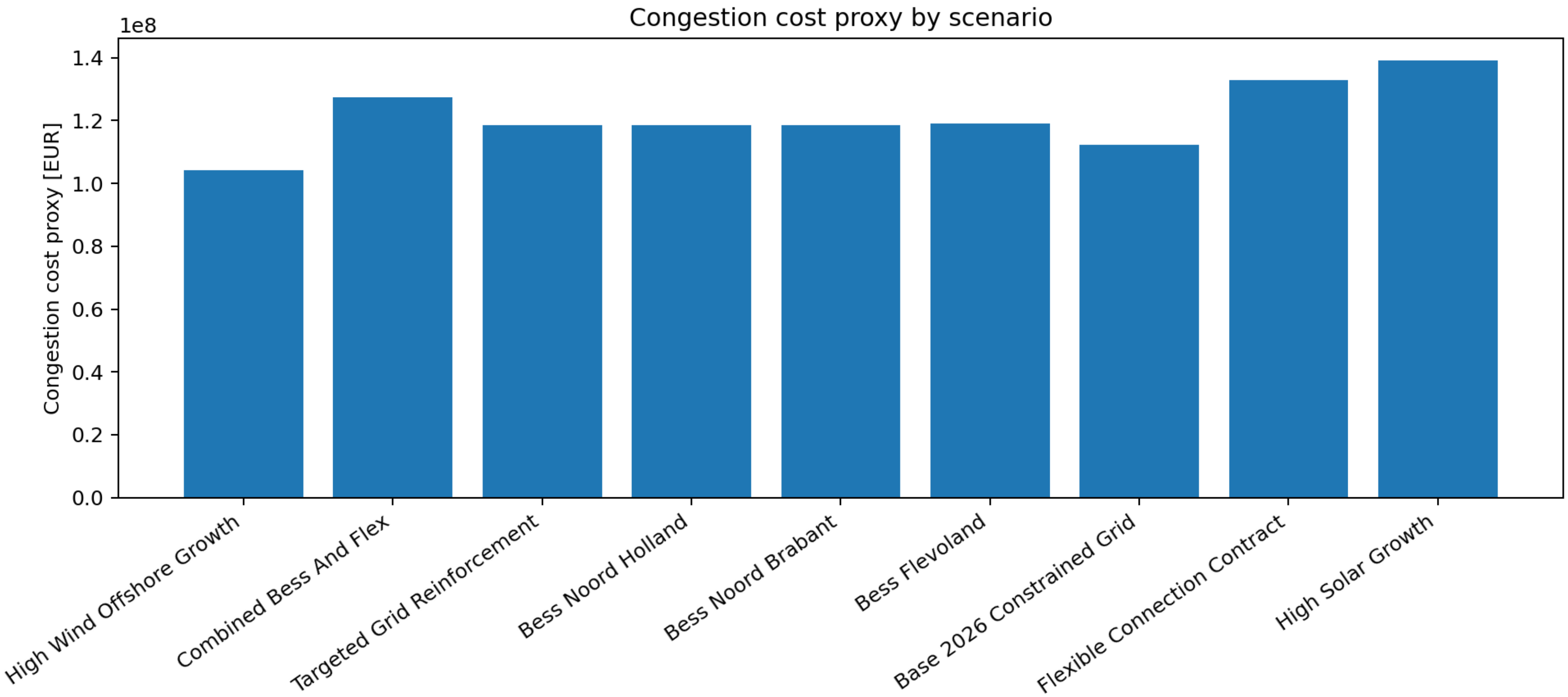
Bess Congestion Value Per Mwh



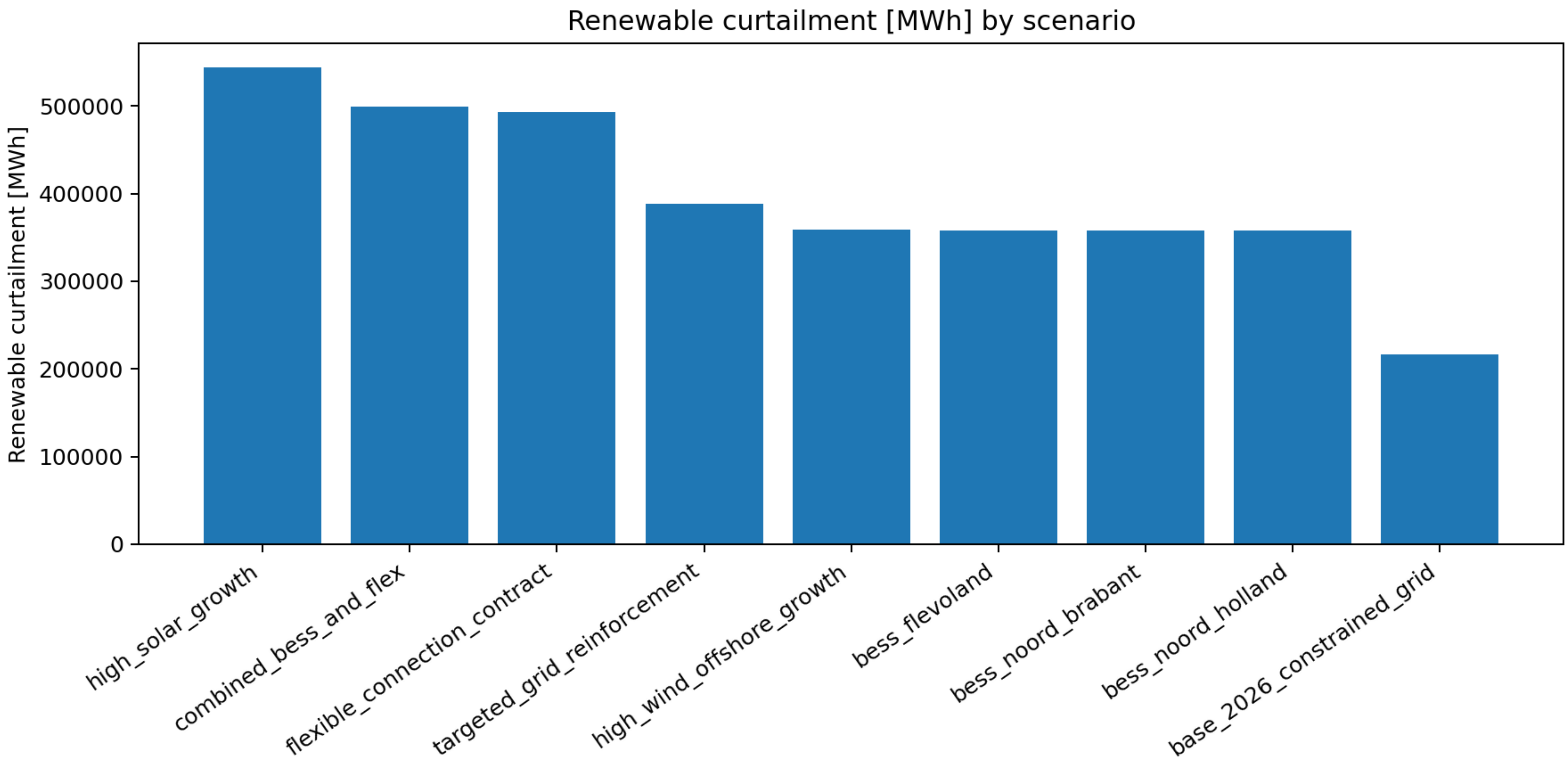
Bess Siting Sizing Score



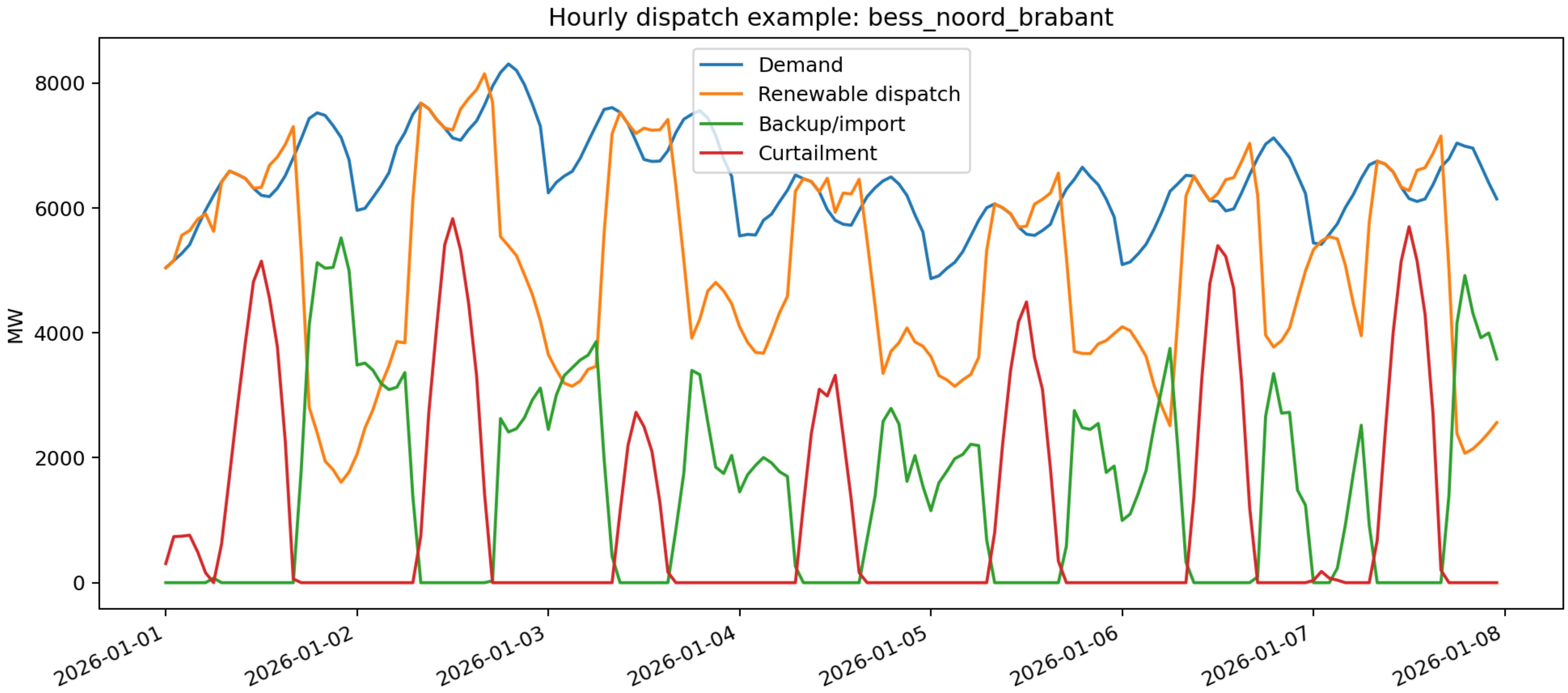
Congestion Cost Proxy By Scenario



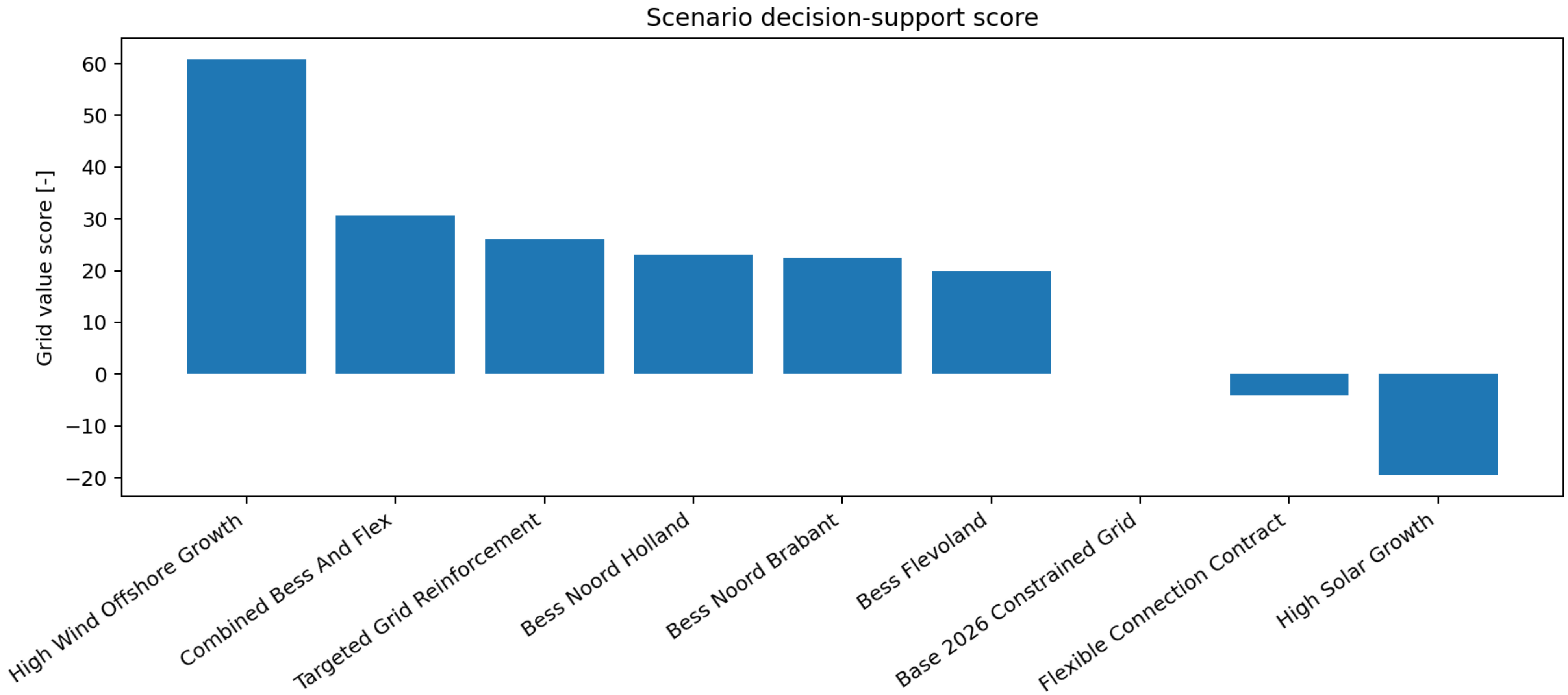
Curtailment By Scenario



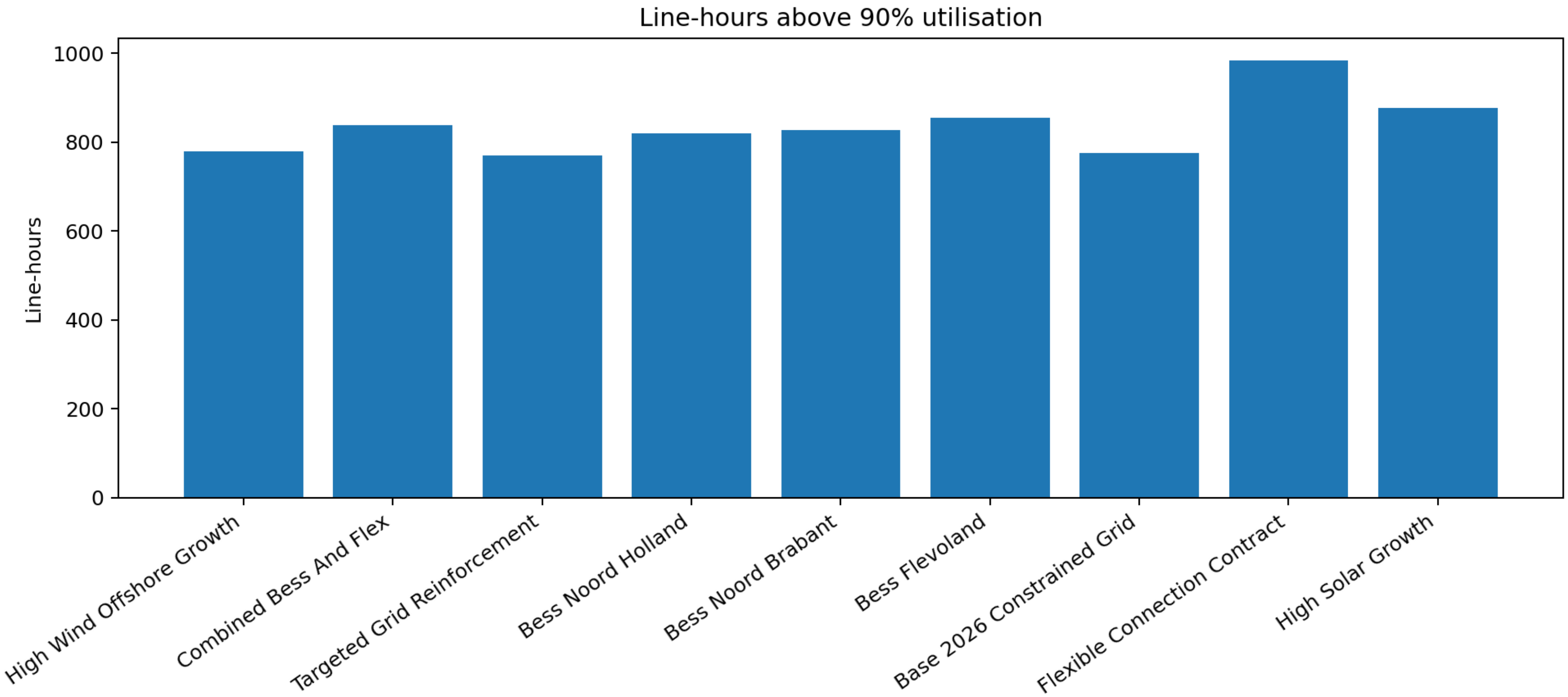
Dispatch Example



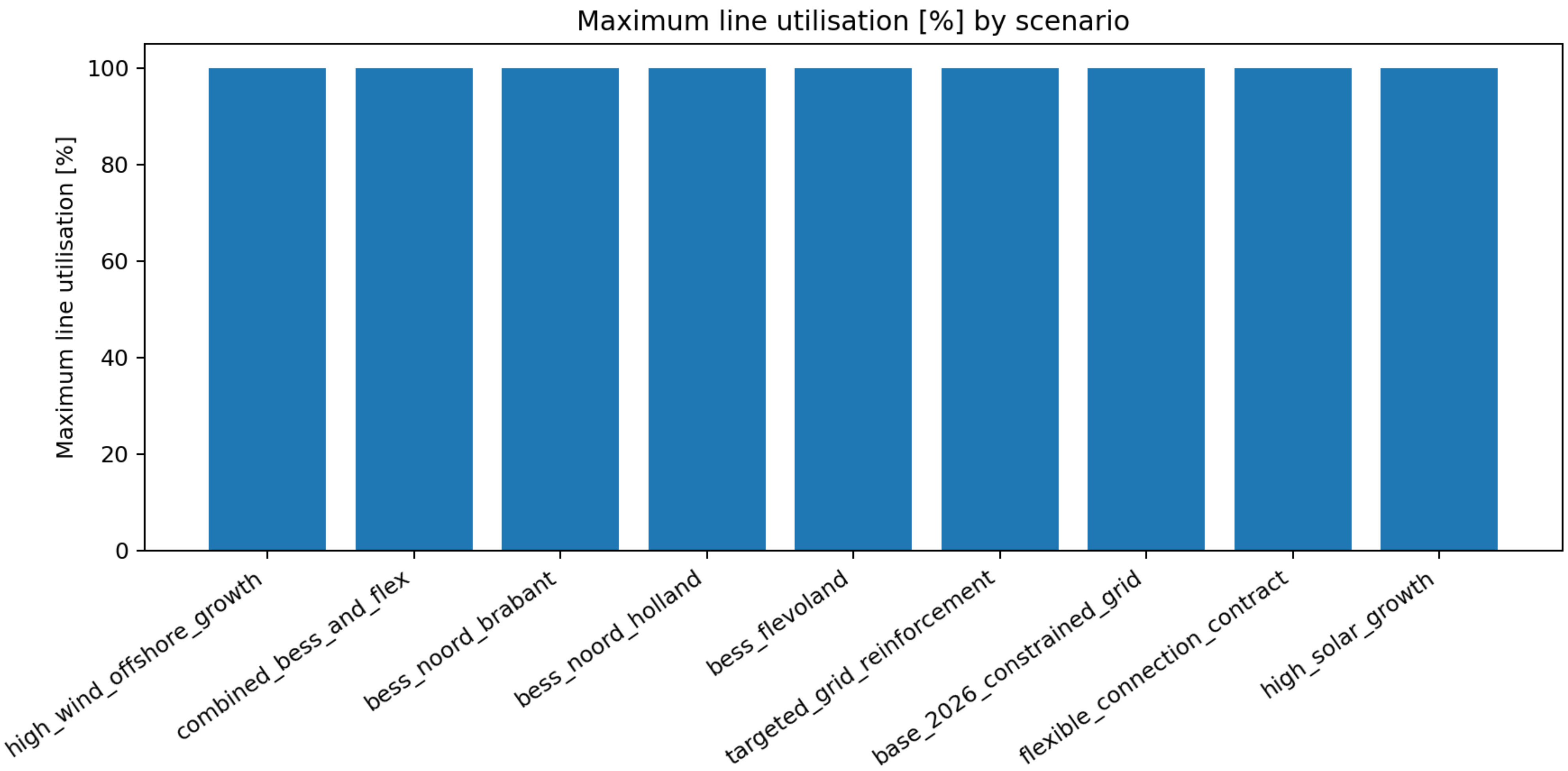
Grid Value Score By Scenario



Line Overload Hours By Scenario

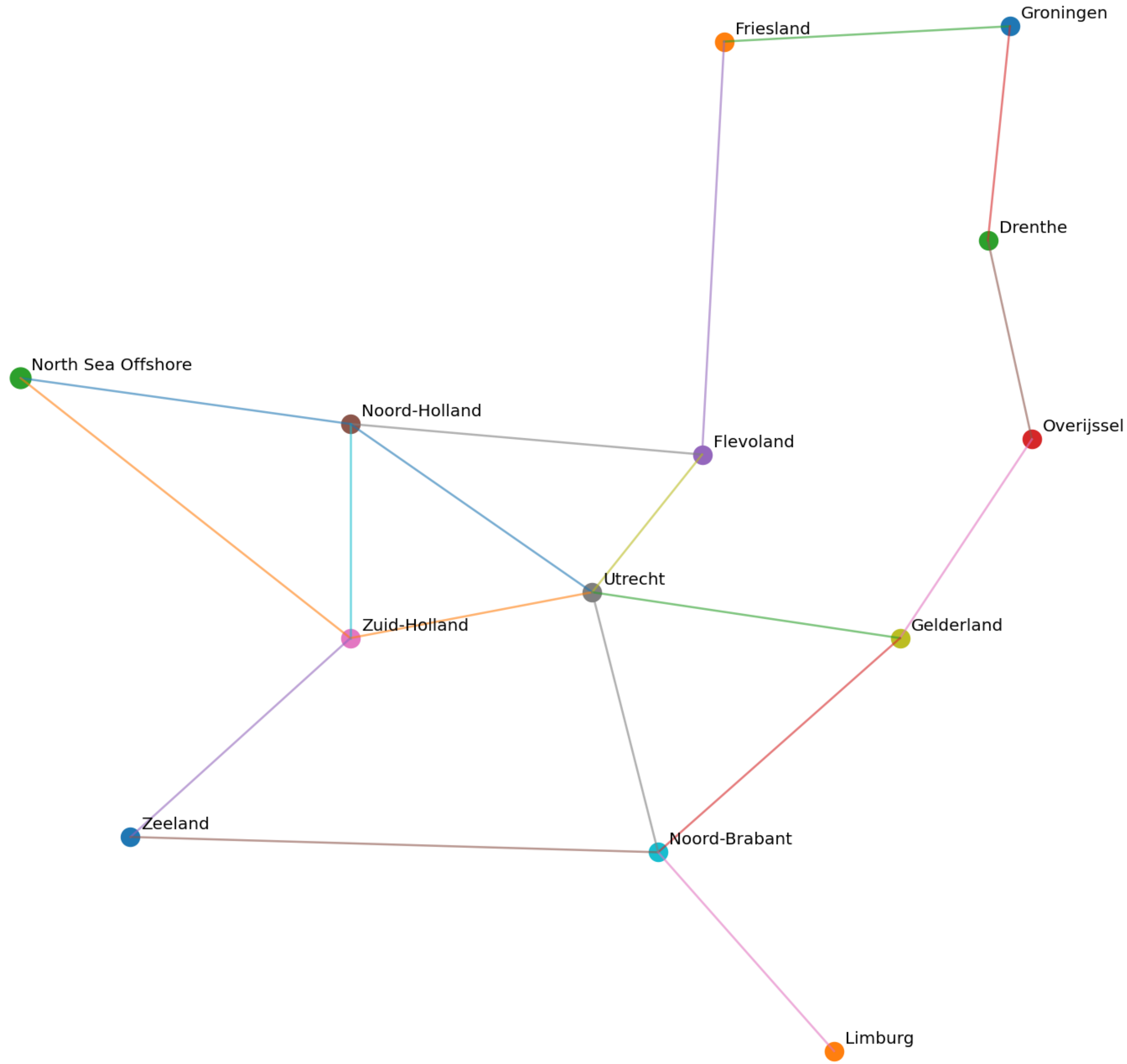


Max Line Utilisation By Scenario

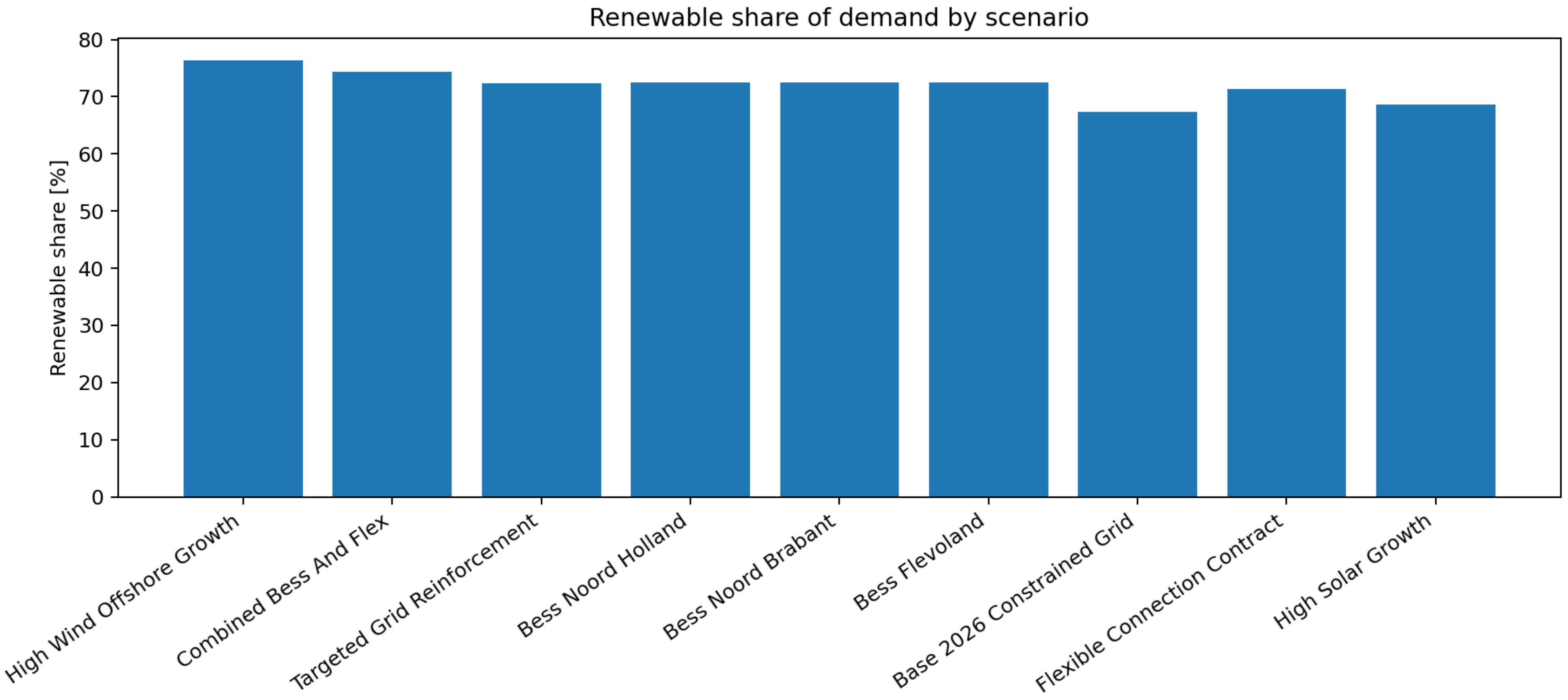


Network Diagram

Synthetic Netherlands-inspired regional grid



Renewable Share By Scenario



Generated report files

Markdown and narrative reports generated by the pipeline

report	purpose	size_bytes
bess_siting_report.md	Bess Siting Report	1,421.00
executive_grid_flexibility_report.md	Executive Grid Flexibility Report	2,889.00
portfolio_summary.md	Portfolio Summary	2,354.00
scenario_report.md	Scenario Report	4,100.00

The full narrative content remains available in the repository outputs folder and in the embedded ZIP bundle.

File inventory

Files included in the downloadable bundle

category	file	size_bytes
config	config/model_config.yaml	3,523.00
config	config/scenarios.yaml	3,216.00
figures	outputs/figures/backup_dispatch_by_scenario.png	132,693.00
figures	outputs/figures/bess_congestion_value_per_mwh.png	105,147.00
figures	outputs/figures/bess_siting_sizing_score.png	109,528.00
figures	outputs/figures/congestion_cost_proxy_by_scenario.png	131,562.00
figures	outputs/figures/curtailment_by_scenario.png	132,765.00
figures	outputs/figures/dispatch_example.png	253,757.00
figures	outputs/figures/grid_value_score_by_scenario.png	125,267.00
figures	outputs/figures/line_overload_hours_by_scenario.png	122,441.00
figures	outputs/figures/max_line_utilisation_by_scenario.png	122,115.00
figures	outputs/figures/network_diagram.png	121,600.00
figures	outputs/figures/renewable_share_by_scenario.png	129,278.00
inputs	README.md	8,224.00
inputs	pyproject.toml	592.00
inputs	requirements.txt	118.00
processed data	data/processed/offshore_wind_profile.csv	6,755.00
processed data	data/processed/regional_demand_profiles.csv	40,453.00
processed data	data/processed/regional_solar_profiles.csv	27,275.00
processed data	data/processed/regional_wind_profiles.csv	42,431.00
raw data	data/raw/README.md	1,919.00
reports	outputs/reports/bess_siting_report.md	1,421.00

category	file	size_bytes
reports	outputs/reports/executive_grid_flexibility_report.md	2,889.00
reports	outputs/reports/portfolio_summary.md	2,354.00
reports	outputs/reports/scenario_report.md	4,100.00
tables	outputs/tables/bess_business_case.csv	49,597.00
tables	outputs/tables/bess_siting_sizing_sweep.csv	40,694.00
tables	outputs/tables/bess_top10_siting_sizing_options.csv	6,376.00
tables	outputs/tables/bottleneck_diagnostics.csv	20,682.00
tables	outputs/tables/hourly_dispatch.csv	2,918,529.00
tables	outputs/tables/hourly_results.csv	2,920,012.00
tables	outputs/tables/n1_security_proxy.csv	29,648.00
tables	outputs/tables/scenario_summary.csv	5,632.00
tables	outputs/tables/solve_log.csv	339.00
tables	outputs/tables/validation_summary.csv	933.00

The full machine-readable bundle is attached to this PDF as **pypsa_nl_grid_flexibility_report_bundle.zip**. That attachment contains the complete input, output, figure and report set.